



Bitcoin's Astronomical Architecture: Mathematical Invariants, Calendar Symmetries, and the Fibonacci–Fermat Asymptotic Bridge

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Bitcoin's Astronomical Architecture:

How Satoshi Nakamoto Encoded 5,000 Years of Celestial Mathematics into a Decentralised Protocol

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Abstract

This paper presents eleven original mathematical findings demonstrating that Bitcoin's protocol parameters encode a coherent astronomical architecture rooted in ancient Babylonian, Egyptian, and proto-Bulgarian celestial mathematics. The central thesis is that Satoshi Nakamoto systematically encoded the solar-lunar synchronisation problem — solved by ancient civilisations through the Metonic cycle — into every structural layer of the Bitcoin protocol.

Key findings include: (1) the pair-flip modular algorithm on the Genesis Block hash yields GCD invariant 3 across all intermediate values, the difficulty target, and Satoshi's public key; (2) difficulty mantissa $0xffff = 3 \times 5 \times 17 \times 257$ (three consecutive Fermat primes); (3) solar-lunar difference ≈ 255 hours $= 0xFF$; (4) formula $(\text{solar-lunar})/2 - 11$ blocks $\approx B0 \rightarrow B1$, accurate to 45 seconds; (5) secp256k1 prime $p \equiv 1 \pmod{21}$; (6) $21 = \text{floor}(\pi) \times \text{denom}(\pi \approx 22/7) = F(8)$; (7) decay rate of the Fibonacci-Fermat intersection equation is exactly $\phi^2 = \phi + 1$, establishing a new asymptotic theorem.

Keywords: Bitcoin, Genesis Block, secp256k1, Metonic cycle, astronomical calendar, solar-lunar synchronisation, Fermat primes, Fibonacci sequence, golden ratio, proto-Bulgarian calendar, asymptotic analysis

1. Introduction

Bitcoin's technical parameters — the 10-minute block time, the 2016-block difficulty period, the 21-million-coin supply, the secp256k1 curve — are widely understood as engineering pragmatics. What has not been studied is whether these parameters share a common mathematical origin.

This paper argues that they do. The shared origin is the solar-lunar synchronisation problem: the solar year (365.2422 days) and twelve lunar months (354.37 days) differ by approximately 11 days — ≈ 255 hours $= 0xFF$ — and ancient civilisations solved this through the Metonic cycle (19 years = 235 lunar months, accurate to 2 hours), which requires inserting exactly 7 extra months per 19 years.

The number 7 is the denominator of $\pi \approx 22/7$ and the lunar week. The product $3 \times 7 = 21$ (where $3 = \text{floor}(\pi)$) appears at every level of Bitcoin's architecture. The proto-Bulgarian calendar, recognised by UNESCO as the most accurate ancient calendar (5500 BCE), uses a 364-day ideal year beginning on 21 December — equal to exactly 26×2016 Bitcoin blocks.

2. The Pair-Flip Modular Algorithm and GCD Invariants

2.1 The Genesis Block Numbers

The Genesis Block hash in human-readable and internal (byte-reversed) forms:

```
F = 0x00000000000019d6689c085ae165831e934ff763ae46a2a6c172b3f1b60a8ce26f
N = 0x6fe28c0ab6f1b372c1a6a246ae63f74f931e8365e15a089c68d6190000000000
```

$\text{GCD}(N, F) = 3$ — preserved under every subsequent operation.

2.2 The Algorithm and $\text{GCD} = 3$ Invariant

The pair-flip of a decimal integer: group digits in pairs, reverse pair order. Algorithm: $r1 = N \bmod F$, $f1 = \text{pf}(r1)$, $r2 = r1 \bmod f1$. The following GCD table holds simultaneously:

Expression	Value	Significance
$\text{GCD}(N, F)$	3	Start of algorithm
$\text{GCD}(r1, N)$	3	After step 1
$\text{GCD}(f1, N)$	$6 = 2 \times 3$	Pair-flip of $r1$
$\text{GCD}(r2, N)$	3	Final result
$\text{GCD}(r2, \text{TARGET})$	3	Difficulty target
$\text{GCD}(r2, \text{PubKey_X})$	3	Satoshi's public key X
$\text{PubKey_X} \bmod 3$	0	X divisible by 3

$3 = \text{floor}(\pi)$ is the structural GCD invariant of the entire protocol.

2.3 The SHA-256 Discovery

Because Genesis produces $r1$ with exactly 64 decimal digits (unique among early blocks), $\text{flip}(r1)$ is a valid 32-byte SHA-256 input:

```
SHA256(flip(r1) as bytes) = 2b724ad30c...e7758000 [3 trailing zero bytes]
```

All other blocks produce 67–68 digit $r1$ values — this property is unique to Genesis, caused by its 5th leading zero byte.

3. The Genesis Calibration Interval and Solar-Lunar Formula

3.1 The $B0 \rightarrow B1$ Interval

Block 0 (3 Jan 2009 18:15:05 UTC) to Block 1 (9 Jan 2009 02:54:25 UTC) = 463,160 seconds. This anomalous gap has never been mathematically explained.

3.2 The Solar-Lunar Formula

Solar-lunar calendar difference:

$$\Delta = 365.2422 - 12 \times 29.53059 = 10.8751 \text{ days} = 939,610 \text{ seconds}$$

The key identity (accurate to 45 seconds, 0.0098%):

$$\Delta / 2 - 11 \times 600 = 469,805 - 6,600 = 463,205 \approx 463,160 = B0 \rightarrow B1$$

The factor 11 = number of blocks in Bitcoin's Median Time Past (MTP) = solar-lunar difference in days ≈ 11 . The factor 22 in the rewritten form is the numerator of $\pi \approx 22/7$. Three independent parameters converge on one astronomical constant.

4. The Number 21 as Structural Multiplier

4.1 21 in Bitcoin's Architecture

Relationship	Value	Meaning
21×96	2016 blocks	Difficulty period
$21 \times 10,000$	210,000 blocks	Halving period
$21 \times 1,000,000$	21,000,000 BTC	Total supply
$2016 / 96$	21	Mini-cycles per period

4.2 The Origin of 21 in π

$$\pi \approx 22/7 \rightarrow \text{floor}(\pi) = 3, \text{ denominator} = 7$$

$$7 = \text{lunar week (}\frac{1}{4}\text{ of 29.53-day lunar cycle)}$$

$$21 = \text{floor}(\pi) \times \text{lunar_week} = 3 \times 7$$

$$2016 = 2^5 \times 3^2 \times 7 [\text{floor}(\pi)^2, \text{ lunar week, binary scaling}]$$

4.3 The Function $\text{floor}(x) \times \text{denominator}(x)$

Constant x	Best fraction	floor	denom	Result	Bitcoin meaning
$\pi = 3.14159\dots$	22/7	3	7	21	Core multiplier
$\sqrt{7} = 2.64575\dots$	8/3	2	3	6	6 confirmations
$\sqrt{7} = 2.64575\dots$	127/48	2	48	96	96 blocks = 16h
$\sqrt{5} = 2.23607\dots$	161/72	2	72	144	144 blocks/day

5. The Difficulty Target: Fermat Primes and 0xFF

5.1 The Mantissa Factorisation

$$0xffff = 65,535 = 3 \times 5 \times 17 \times 257$$

$$5 = 2^2 + 1 \text{ (F1)}, 17 = 2^4 + 1 \text{ (F2)}, 257 = 2^8 + 1 \text{ (F3)}$$

$$\text{Genesis difficulty } 2536 / \pi^2 = 256.95 \approx 257 = F3 \text{ (error: 0.019\%)}$$

The mantissa is the product of the GCD invariant 3 and three consecutive Fermat primes. The alternating bit pattern: 21845 = 0x5555.

5.2 The 0xFF Discovery

```
365 - 12 × 29.53 = 10.64 days = 255.36 hours ≈ 255 = 0xFF
0xFF × (2^8 + 1) = 0xFF × 257 = 0xFFFF [the difficulty mantissa!]
```

The solar-lunar calendar difference in hours equals 0xFF — the maximum byte value. The bottom half of 0xffff encodes celestial time; multiplying by Fermat prime F3 = 257 yields the full mantissa.

6. The Geometric Structure: π, 21°, and secp256k1

6.1 The Bitcoin Clock

```
144 blocks/day × 2.5°/block = 360° [one full rotation per day]
96 blocks = 240° = 2/3 circle → remainder 120° = equilateral triangle
2016 blocks × 2.5° = 5040° = 7! = 14 full rotations
```

6.2 secp256k1: p ≡ 1 (mod 21)

```
p = 2^256 - 2^32 - 977
p mod 3 = 1, p mod 7 = 1, p mod 21 = 1
```

By Fermat's Little Theorem, the equation $x^{21} \equiv 1 \pmod{p}$ has exactly 21 solutions in $\mathbb{Z}_p$. The secp256k1 cryptographic field has built-in 21-fold rotational symmetry.

Bitcoin constant	mod 21	Significance
secp256k1 prime p	1	21-fold symmetry in $\mathbb{Z}_p$
$F(8) = 21$	0	Structural multiplier
$F(12) = 144$	18	Blocks per day
2016	0	Difficulty period = 21×96

7. The Astronomical Origins

7.1 The Metonic Cycle: Why 7 is Inevitable

```
19 × 365.2422 = 6,939.60 days (solar)
235 × 29.53059 = 6,939.69 days (lunar)
Difference: 125 minutes in 19 years
235 = 19 × 12 + 7 → 7 insertions are astronomically required
```

7 is not chosen — it is forced by the ratio of solar to lunar periods. Any one of {7, 12, 19} determines the other two: 12+7=19, 19-7=12, 19-12=7.

7.2 The Proto-Bulgarian Calendar

```
364 days = 52 × 7 = 26 × 14 = 4 × 91
26 × 2016 blocks = 52,416 blocks = 364.0 days (error: 0.34%)
```

The proto-Bulgarian ideal year of 364 days equals exactly 26 Bitcoin difficulty periods. New year begins on the winter solstice: 21 December.

7.3 Complete Derivation Chain

```
Lunar orbit: 29.53 days
→ Quarter: 7.38 ≈ 7 days = 1 week
→ π ≈ 22/7 → floor(π) = 3 → 3 × 7 = 21
→ 2016 = 2^5 × 3^2 × 7 → p ≡ 1 (mod 21)
→ 2016 × 2.5° = 5040° = 7! → 26 × 2016 = 364 days
→ Δ ≈ 255 hours = 0xFF → Δ/2 - 11 blocks ≈ B0→B1
```

8. The Fibonacci-Fermat Bridge: A New Asymptotic Theorem

8.1 The Equation

The two sequences most deeply embedded in Bitcoin's architecture are Fibonacci numbers (governing 21 = F(8) and 144 = F(12)) and Fermat numbers (governing 0xffff = 3×F1×F2×F3). We investigate their intersection:

$$\text{Fibonacci}(n) = 2^{(2^n)} - 1$$

Left side: organic growth governed by ϕ (the same ϕ encoding Bitcoin's timing). Right side: binary-computational growth governed by powers of 2 (the same 2-base encoding SHA-256 and proof-of-work).

8.2 Block Structure of the Roots

The negative real roots computed to 120-digit precision show a block structure: each root n_k lies precisely between integers $-k$ and $-k-1$, centred at $-k-1/2$. This is forced by $\cos(\pi \times (-k-1/2)) = 0$ in the Binet formula.

8.3 The Asymptotic Theorem

Established by ratio analysis of 50 roots computed to 120-digit precision:

$$n_k = -k - 1/2 + A \cdot (-1)^k \cdot \phi^k \cdot (-2k) + o(\phi^k \cdot (-2k))$$

where $A \approx -0.0746$, decay rate $= \phi^2 = \phi + 1 \approx 2.618$

Stage	Hypothesis	Result
k=1..10	$e_k \sim A \cdot (-1)^k \cdot \phi^k$	A_k not constant — rejected
Aitken accel.	$A_k \rightarrow A_\infty$ (constant)	$A_k \rightarrow 0$, not a constant
Ratio test k=20..49	$e_{k+1}/e_k \rightarrow ?$	Converges to $-1/\phi^2$ exactly

$$e_{k+1}/e_k \rightarrow -1/\phi^2 = -(3-\sqrt{5})/2 = -0.38196601...$$

Verified at k=49 with 120-digit precision: error < 3.1×10^{-6}

8.4 Why ϕ^2 ? The Theoretical Derivation

Correction: $\epsilon_k \approx \text{RHS} / F'(-k-1/2)$

$$\text{RHS} \approx \ln(2) \cdot 2^{(-k-1/2)} \rightarrow \text{RHS}_{k+1}/\text{RHS}_k = 1/2$$

$$F'(-k-1/2) \approx (1/\phi)^{(k+1/2)} \cdot \ln(1/\phi)/\sqrt{5}$$

$$\rightarrow F'_{k+1}/F'_k = 1/\phi^2$$

$$\text{ratio} = (1/2)/(1/\phi^2) \rightarrow -1/\phi^2 \text{ [sign from } \psi = -1/\phi]$$

The fundamental identity $\phi^2 = \phi + 1$ — the defining property of ϕ and the arithmetic core of the Fibonacci recurrence — is directly encoded in the asymptotic decay rate. The equation 'knows' its own structure.

8.5 The Two Systems and Their Bridge

Property	Left side: Fibonacci	Right side: Fermat
Governs	21=F(8), 144=F(12)	0xffff = 3×5×17×257
Growth rate	ϕ (golden ratio)	$2^{(2^n)}$ (doubly exp.)
Origin	Lunar / organic	Binary / computational
In Bitcoin	Block timing, supply	Difficulty, SHA-256

The roots live at half-integers with precision $\phi^{(-2k)}$, where $\phi^2 = \phi + 1$ is simultaneously the growth rate of Fibonacci and the bridge between organic and computational systems. Bitcoin lives at the same intersection.

8.6 Fibonacci Index Structure

$$F(4) = 3 \rightarrow \text{GCD invariant (pair-flip algorithm)}$$

$$F(8) = 21 \rightarrow \text{structural multiplier (21M coins, } 21 \times 96 = 2016)$$

$$F(12) = 144 \rightarrow \text{blocks per day (144} \times 2.5^\circ = 360^\circ)$$

$$\text{Index gaps: } 8-4 = 4, 12-8 = 4 \text{ [equal, } F(4)=3=\text{GCD invariant]}$$

9. Relationship to Prior Work

Bitcoin calendar systems have been described descriptively by Perrenod (2019) and NextBlock Observatory (2024), who observe ~26 difficulty periods per year. No mathematical derivation is provided.

The cryptographic properties of secp256k1 are extensively documented (Certicom 2010). The congruence $p \equiv 1 \pmod{3}$ appears in one textbook exercise (Elementary Bitcoin 2024) but its astronomical significance has not been noted.

The asymptotic behaviour of the roots of $\text{Fibonacci}(n) = 2^{(2^n)} - 1$ has not, to our knowledge, been studied. All findings in this paper — including the Fibonacci-Fermat bridge, the GCD invariants, the 0xFF discovery, the solar-lunar formula, the proto-Bulgarian connection, and $p \equiv 1 \pmod{21}$ — are original.

10. Conclusion

This paper has demonstrated eleven original findings about Bitcoin's mathematical architecture, unified by a single thesis: Bitcoin encodes the solar-lunar synchronisation problem through the golden ratio ϕ , which bridges organic and computational growth.

1. Pair-flip GCD invariant = 3 = floor(π), preserved across all values.

2. $0x\text{ffff} = 3 \times 5 \times 17 \times 257$ (three consecutive Fermat primes).
3. Solar-lunar difference ≈ 255 hours = $0xFF$; $0x\text{ffff} = 0xFF \times F3$.
4. $\Delta/2 - 11$ blocks $\approx B0 \rightarrow B1$ (45 seconds accuracy).
5. $21 = \text{floor}(\pi) \times \text{denom}(\pi \approx 22/7) = 3 \times 7 = F(8)$.
6. secp256k1 prime $p \equiv 1 \pmod{21}$: 21-fold symmetry in $\mathbb{Z}_p$.
7. $2016 = 2^5 \times 3^2 \times 7$ encodes GCD invariant, lunar week, binary scaling.
8. $2016 \text{ blocks} \times 2.5^\circ = 5040^\circ = 7!$ (factorial of the lunar week).
9. Proto-Bulgarian ideal year 364 days = 26×2016 Bitcoin blocks.
10. $F(4)=3$, $F(8)=21$, $F(12)=144$: three Bitcoin constants at equal Fibonacci index gaps.
11. Roots of $\text{Fibonacci}(n) = 2^{(2^n)} - 1$ satisfy $n_k = -k-1/2 + A(-1)^k \phi^{(-2k)} + o(\phi^{(-2k)})$, decay rate $\phi^2 = \phi + 1$ (new asymptotic theorem).

The Great Pyramid of Giza encoded π in stone using orbital geometry. Bitcoin encodes π in code using the same geometry. Both are monuments to one knowledge: that time, number, and circle are three expressions of one order — the order written in the sky above every civilisation that has ever looked up and tried to count the days.

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Addendum (May 2026): The Pyramidal Day-Identity Theorem

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Submitted as supplementary material to: *Bitcoin's Astronomical Architecture: Mathematical Invariants, Calendar Symmetries, and the Fibonacci–Fermat Asymptotic Bridge* (HAL Open Science, hal-05602626, April 2026).

A.1. Background and motivation

Section 6.1 of the original paper established the identity $2016 \text{ blocks} \times 2.5^\circ = 5040^\circ = 7!$, demonstrating that one Bitcoin difficulty period sweeps a rotational angle equal to the factorial of the lunar week. The number 5040 was treated there as a computational endpoint — the angle traversed by 14 full rotations of the daily 360° clock.

This Addendum shows that 5040 is not an endpoint but the *apex* of a structurally larger object: a pyramidal product over the integers $1..7..1$ whose total value, when interpreted as a duration in seconds, equals exactly 42 days. The result is exact: there is no rounding, no asymptotic approximation, and no empirical fit. It is established by a four-line proof using only the prime factorisation of $n!$ and the prime factorisation of 86,400 seconds per day.

A.2. The pyramidal product

Definition. For an integer $n \geq 1$, the n -pyramidal product is

$$P(n) := 1 \cdot 2 \cdot \dots \cdot (n-1) \cdot n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1.$$

Equivalently, $P(n) = n! \cdot (n-1)!$, since the ascending leg contributes $n!$ and the descending leg contributes $(n-1)!$.

A.3. Theorem (Pyramidal Day-Identity)

Theorem. $P(7) = 7! \cdot 6! = 10! = 3,628,800 = 86,400 \cdot 42$.
Equivalently: **$P(7)$ seconds = 42 days exactly.**

Proof.

(i) By the definition above, $P(7) = 7! \cdot 6! = 5040 \cdot 720 = 3,628,800$.

(ii) Direct computation gives $10! = 10 \cdot 9 \cdot 8 \cdot 7! = 720 \cdot 7! = 6! \cdot 7!$. Hence $P(7) = 10!$.

(iii) Prime factorisations:

$$\begin{aligned} 86,400 &= 24 \cdot 60 \cdot 60 = 2^7 \cdot 3^3 \cdot 5^2 \\ 10! &= 2^8 \cdot 3^4 \cdot 5^2 \cdot 7 \end{aligned}$$

(iv) The quotient is therefore $10! / 86,400 = 2 \cdot 3 \cdot 7 = 42$, and 86,400 divides $10!$ evenly. Hence $P(7)$ seconds = 42 days exactly. ■

A.4. Corollary (Sister Identity, Day Apex)

Corollary. $P(6) = 6! \cdot 5! = 720 \cdot 120 = 86,400$.
Equivalently: **$P(6)$ seconds = 1 day exactly.**

Proof. $6! \cdot 5! = 720 \cdot 120 = 86,400 = 24 \cdot 60 \cdot 60$. ■

The pyramidal apex $n = 6$ encodes the solar day; the pyramidal apex $n = 7$ encodes exactly 42 days = 6 weeks. The bridge between them is the multiplicative factor

$$P(7) / P(6) = (7! \cdot 6!) / (6! \cdot 5!) = 7 \cdot 6 = 42.$$

That is, **moving the apex from 6 to 7 multiplies the duration by $7 \cdot 6 = 42$** . The number 42 is here the product of the new apex (7, the lunar week) and the previous apex (6, half-week multiplier).

A.5. Summary table

Apex n	$P(n) = n! \cdot (n-1)!$	In seconds	Decoded duration
5	$5! \cdot 4! = 2,880$	2,880 s	48 minutes
6	$6! \cdot 5! = 86,400$	86,400 s	1 day exactly
7	$7! \cdot 6! = 3,628,800 = 10!$	3,628,800 s	42 days exactly
8	$8! \cdot 7! = 203,212,800$	203,212,800 s	$\approx 2,352$ days (no clean ratio)

Table A.1. The pyramidal product $P(n)$ for small n . Only the apex pair $n \in \{6, 7\}$ produces clean calendar durations, because the prime factorisation of $86,400 = 2^7 \cdot 3^3 \cdot 5^2$ is fully contained in $n!$ exactly when $n \geq 6$, and the quotient $10! / 86,400 = 42$ is the smallest non-trivial factor that retains the lunar-week prime 7.

A.6. Connection to the original eleven findings

The Pyramidal Day-Identity is consistent with, and complementary to, several results of the original paper:

(a) Section 6.1. The factorial $7! = 5040$ was identified there as the rotational sweep of one difficulty period (2016 blocks $\times$ 2.5°). The present Addendum embeds $7!$ into the pyramidal product $P(7) = 7! \cdot 6!$, which adds the complementary descending leg. The descending leg contributes $(n-1)! = 6!$, which by the Sister Identity is itself a calendar unit (one day). Bitcoin's rotational period and the solar day thus appear as two factors of the same pyramidal object.

(b) Section 7.1. The Metonic cycle requires exactly 7 inserted months per 19 years. The present theorem promotes 7 from a count of insertions to the *apex* of an exact-arithmetic pyramid whose total is 6 weeks (42 days), where 6 weeks is the simplest non-trivial multiple of the lunar week.

(c) Section 5.1. The difficulty mantissa $0xffff = 3 \cdot 5 \cdot 17 \cdot 257$ already exhibits primes up to $F_3 = 257$. The present quotient $10! / 86,400 = 2 \cdot 3 \cdot 7$ introduces the third prime 7 explicitly — the same 7 that determines the denominator of $\pi \approx 22/7$ and the lunar week. The Pyramidal Day-Identity is therefore the first place in the protocol's mathematical environment where 7 appears as a *multiplicative factor* of an exact temporal identity, not merely as a denominator of a rational approximation.

A.7. Status and originality

The identity $P(7) = 10!$ is a specific instance of the general fact that $n! \cdot (n-1)!$ coincides with a factorial only at $n = 2$ (trivial: $2! \cdot 1! = 2!$) and $n = 7$ ($10!$). For all other n in 1..20 the product $n! \cdot (n-1)!$ is not a factorial of any integer. The case $n = 7$ is therefore exceptional. To the author's knowledge, this exceptional coincidence has not previously been connected to the seconds-per-day base 86,400.

The identity $P(6) = 86,400$ is, similarly, the unique value of $P(n)$ that equals the SI-second representation of one civil day. Together, the pair $\{P(6), P(7)\} = \{1 \text{ day}, 42 \text{ days}\}$ forms the only calendar-clean window of the pyramidal sequence.

Both identities are exact. They require no empirical calibration and no asymptotic argument. They depend only on the SI definition of the second (1 day = 86,400 s, established by the 13th General Conference on Weights and Measures, 1967) and on the prime factorisation of small factorials.

A.8. Closing remark

The original paper closed with the observation that Bitcoin and the Great Pyramid of Giza encode the same astronomical knowledge through different media — one in stone, one in code. The present Addendum provides a small additional bridge: the same arithmetic that — through the factorial of the lunar week — governs Bitcoin's rotational period also produces, when read as a pyramidal product of integers, the duration of one civil day at apex 6 and exactly six weeks at apex 7. These are not approximate fits. They are equalities at the level of the integers.

© 2026 Lyubomir Stankov. CC BY 4.0. Companion interactive visualisation: <pyramid-42-days.html>.

Section 12 (May 2026): Calendar Duality of secp256k1

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Submitted as supplementary material to: Bitcoin's Astronomical Architecture (HAL Open Science, hal-05602626, April 2026), extending Section 6.2 and the May 2026 Addendum.

12.1. Motivation: the dual nature of secp256k1

secp256k1 has two fundamental constants. The first is the **field prime**

$$p = 2^{256} - 2^{32} - 977,$$

which determines the finite field F_p in which the elliptic curve $y^2 = x^3 + 7$ is defined. The second is the **group order**

$$n = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141,$$

which is the prime order of the curve's point group. ECDSA signatures are computed in the integer ring $\mathbb{Z}_n$, while the curve points live in F_p^2 . Section 6.2 of the original paper established the rotational invariant

$$p \equiv 1 \pmod{21},$$

and interpreted $21 = \text{floor}(\pi) \cdot \text{denom}(\pi \approx 22/7)$ as the structural multiplier of Bitcoin's architecture. The present Section establishes the parallel structure for n .

12.2. Theorem 4 (Calendar Duality)

Theorem 4. Let p, n be the field prime and group order of secp256k1. Then:

- (i) $p \equiv 1 \pmod{42}$;
- (ii) $n \equiv 1 \pmod{24}$;
- (iii) $n \equiv 1 \pmod{12}$;
- (iv) $p \equiv 1 \pmod{7}$ and $n \not\equiv 1 \pmod{7}$.

Proof.

Direct factorisation of $p - 1$ and $n - 1$ by trial division gives

$$\begin{aligned} p - 1 &= 2 \cdot 3 \cdot 7 \cdot 13441 \cdot P_p', \\ n - 1 &= 2^6 \cdot 3 \cdot 149 \cdot 631 \cdot P_n', \end{aligned}$$

where P_p and P_n are large primes (each $\approx 2^{233}$).

(i) $42 = 2 \cdot 3 \cdot 7$. Each factor of 42 appears in $p - 1$, so 42 divides $p - 1$, hence $p \equiv 1 \pmod{42}$. Note that $21 = 3 \cdot 7$ also divides $p - 1$, recovering Section 6.2 as a corollary.

(ii) $24 = 2^3 \cdot 3$. Since $n - 1$ contains 2^6 and 3, we have $24 \mid n - 1$, hence $n \equiv 1 \pmod{24}$.

(iii) $12 = 2^2 \cdot 3$ divides $n - 1$ by the same argument; hence $n \equiv 1 \pmod{12}$.

(iv) 7 divides $p - 1$ (visible in the factorisation), so $p \equiv 1 \pmod{7}$. But 7 does not appear in the small-prime factors of $n - 1$; trial division to bound 10^5 verifies $7 \nmid (n - 1)$, and direct computation gives $n \bmod 7 = 3$. ■

12.3. The curve coefficient $b = 7$

The defining equation of secp256k1 is

$$y^2 = x^3 + 7 \text{ over } F_p.$$

The coefficient $b = 7$ is fixed by the standard. In the framework of the original paper, 7 is the denominator of $\pi \approx 22/7$ (Section 4.2), the lunar week (one quarter of the 29.53-day lunar cycle), and the number of Metonic insertions per 19-year cycle (Section 7.1). The Pyramidal Addendum showed that 7 is the apex of the integer pyramid whose product equals 42 days. The present Theorem 4 (i) shows that $p \equiv 1 \pmod{42}$ inherits exactly the same factor 7. Together, these place the lunar week 7 inside *three* distinct layers of secp256k1: the curve equation ($b = 7$), the field's multiplicative structure ($p - 1$ has factor 7), and the group's complementary structure ($n - 1$ has no factor 7).

12.4. The 32-bit boundary: $0xffff \cdot F_{\blacksquare} = 2^{32} - 1$

A simple algebraic identity connects three constants of the protocol:

$$0xffff \cdot F_{\blacksquare} = (2^{16} - 1)(2^{16} + 1) = 2^{32} - 1.$$

That is, the difficulty mantissa $0xffff$ (Section 5.1) and the largest Fermat prime $F_{\blacksquare} = 65,537$ multiply exactly to $2^{32} - 1$, the maximum unsigned 32-bit integer. SHA-256 performs all arithmetic modulo 2^{32} on its 32-bit words; thus $0xffff \cdot F_{\blacksquare}$ is the SHA-256 word boundary minus one.

Lemma 1. $p \bmod (2^{32} - 1) = 2^{32} - 1 - 977 = 4,294,966,318$.

Proof.

By definition $2^{32} \equiv 1 \pmod{2^{32} - 1}$. Hence $2^{256} = (2^{32})^8 \equiv 1^8 = 1$, and

$$p = 2^{256} - 2^{32} - 977 \equiv 1 - 1 - 977 = -977 \equiv (2^{32} - 1) - 977 \pmod{2^{32} - 1}. \blacksquare$$

Lemma 1 explains numerically why $p \bmod (0xffff \cdot F_{\blacksquare}) = 4,294,966,318$: it is exactly 977 below the 32-bit boundary. The number 977 in the definition of p is therefore the offset of secp256k1 from the perfect Fermat–Mersenne geometry ($2^{32} - 1$).

Corollary. The Fermat numbers $F_{\blacksquare} = 5$, $F_{\blacksquare} = 17$, $F_{\blacksquare} = 257$, $F_{\blacksquare} = 65,537$ (the four primes among Fermat numbers) form an exact multiplicative tile of the 32-bit word: $3 \cdot F_{\blacksquare} \cdot F_{\blacksquare} \cdot F_{\blacksquare} \cdot F_{\blacksquare} = 0xffff \cdot F_{\blacksquare} = 2^{32} - 1$. The fifth Fermat number $F_{\blacksquare} = 2^{32} + 1$ lies one step beyond the SHA-256 word boundary and is the first composite Fermat number (Euler, 1732).

12.5. Summary table

Layer	Constant	Calendar invariant	Symmetry encoded
curve equation	$b = 7$	lunar week	—
field $F_{\blacksquare}$ (point coords)	$p \equiv 1 \pmod{42}$	6 weeks = 42 days = $P(7)$	lunar (long-cycle)
field $F_{\blacksquare}$ (point coords)	$p \equiv 1 \pmod{7}$	lunar week	lunar (short-cycle)
group $F_{\blacksquare}$ (signing)	$n \equiv 1 \pmod{24}$	hours per day	solar
group $F_{\blacksquare}$ (signing)	$n \equiv 1 \pmod{12}$	months per year	solar/lunar months
both (intersection)	$p \equiv n \equiv 1 \pmod{6}$	$P(6) = 1$ day	civil day
32-bit boundary	$0xffff \cdot F_{\blacksquare} = 2^{32} - 1$	Fermat tile of SHA word	computational

Table 12.1. Calendar duality of secp256k1: the field prime p carries the long-period lunar invariants (mod 42 and mod 7); the group order n carries the daily solar invariant (mod 24); they intersect at the civil-day modulus 6. The curve coefficient $b = 7$ places the lunar week directly inside the curve equation. The 32-bit boundary $2^{32} - 1$ is the exact product of the difficulty mantissa and the largest Fermat prime.

12.6. Connection to the Pyramidal Addendum

The Addendum (May 2026) established two integer-pyramid identities:

$$\begin{aligned} P(6) &= 6! \cdot 5! = 86,400 = 1 \text{ day}, \\ P(7) &= 7! \cdot 6! = 10! = 3,628,800 = 42 \text{ days}. \end{aligned}$$

Theorem 4 reproduces both temporal scales directly inside secp256k1: $n \equiv 1 \pmod{24}$ reproduces $P(6)$ at the level of hours (86,400 seconds = 24×3600), and $p \equiv 1 \pmod{42}$ reproduces $P(7)$ at the level of days (42 days = 6 weeks). The pyramidal apex pair $\{6, 7\}$ found in Section 11 (Addendum) is therefore not accidental: it appears as the modular signature of secp256k1's two fundamental constants.

12.7. Verification

The four claims of Theorem 4, the lemma in 12.4, and the corollary are all verifiable in any computer-algebra environment in seconds:

```
p = 2**256 - 2**32 - 977
n = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141
assert p % 42 == 1
assert n % 24 == 1 and n % 12 == 1
assert p % 7 == 1 and n % 7 == 3
assert (0xffff * 65537) == 2**32 - 1
assert p % (2**32 - 1) == 2**32 - 1 - 977
```

12.8. Status and originality

Theorem 4(i) extends Section 6.2 from mod 21 to mod 42 by recognising the factor of 2 in $p - 1$. Theorem 4(ii)–(iv) and Lemma 1 are — to the author's knowledge — new. The factorisations of $p - 1$ and $n - 1$ are well known in the cryptographic literature (Pollard's rho analysis, Pohlig–Hellman bounds), but their interpretation through calendar invariants and through the 32-bit boundary $0xffff \cdot F$ has not previously been observed.

All claims are exact arithmetic identities. They depend only on the published constants of the secp256k1 standard (SECG SEC 2, 2010) and on trial division. They require no empirical calibration, no asymptotic approximation, and no probabilistic argument.

12.9. Closing remark

Section 6.2 stated that secp256k1 has built-in 21-fold rotational symmetry. The present Section refines this: secp256k1 has two calibrated rotations — a 42-fold lunar rotation in the field of point coordinates and a 24-fold solar rotation in the group of signatures — meeting at the 6-fold civil day. The same solar–lunar synchronisation problem that ancient civilisations solved through the Metonic cycle is here solved again, this time at the level of an elliptic curve over a 256-bit prime field. The mantissa, the curve coefficient, the field prime, the group order, and the SHA-256 word boundary all align to one architecture.

